Walmart Retail Sales Analytics Dashboard

Project Documentation

Developed by

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Tools and Technologies

MySQL
SQL
Python
Pandas
Microsoft Excel
Power BI
DAX

Project type

Business Intelligence | Data Analytics | Retail Sales Analytics

Date

July 2026

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1. Project Overview

The Walmart Retail Sales Analytics Dashboard is an end-to-end Business Intelligence and Data Analytics project developed to analyze retail sales performance, profitability, customer behavior, product performance, and operational efficiency using MySQL, SQL, Python, Microsoft Excel, Power BI, and DAX.

The project follows a structured analytics workflow beginning with data import and preprocessing in Python. The dataset was examined for data quality issues such as missing values, duplicate records, inconsistencies, and outliers before preparing a cleaned dataset for further analysis. After preprocessing, the cleaned dataset was imported into MySQL, where SQL queries were used to perform business-oriented analysis. The processed data was then connected to Power BI, where a data model, DAX measures, and an interactive dashboard were developed to support business reporting and decision-making.

The final solution consists of a five-page interactive Power BI dashboard designed to analyze business performance from multiple perspectives, including executive KPIs, sales performance, customer behavior, operational efficiency, and strategic business insights.

The project concludes with analytical findings and business recommendations that assist stakeholders in understanding sales trends, profitability patterns, customer purchasing behavior, regional performance, and operational efficiency through interactive visualizations.

2. Business Problem

Retail organizations generate large volumes of transactional data every day through customer purchases, product sales, and shipping operations. Although this data contains valuable business information, it often remains underutilized without systematic analysis and reporting. Decision-makers require timely and accurate insights to understand sales performance, profitability, customer purchasing behavior, product demand, and operational efficiency.

One of the major challenges in retail businesses is evaluating overall business performance beyond sales figures. High sales alone do not necessarily indicate strong business performance, as profitability is influenced by factors such as discounts, shipping costs, product mix, and regional performance. Without proper analysis,

identifying profitable areas, loss-making segments, and operational bottlenecks becomes difficult.

Organizations also require continuous monitoring of key performance indicators (KPIs) such as total sales, profit, orders, customers, and profit margin. Business users need the ability to compare regional performance, evaluate product categories, understand customer segments, and assess shipping operations through interactive reporting rather than static spreadsheets.

The objective of this project is to transform raw retail transaction data into meaningful business insights through data preparation, analytical reporting, and dashboard development. The resulting Business Intelligence solution enables stakeholders to monitor business performance, identify trends, evaluate operational efficiency, and support informed decision-making using interactive visualizations and analytical insights.

3. Project Objectives

The primary objectives of this project are:

- To transform raw retail transaction data into a structured Business Intelligence solution through data preparation, SQL analysis, and interactive dashboard reporting.
- To measure and evaluate overall business performance by analyzing key performance indicators (KPIs) such as Sales, Profit, Orders, Customers, Quantity Sold, and Profit Margin.
- To uncover actionable business insights by identifying sales trends, profitability drivers, customer purchasing behavior, product performance, regional variations, and operational efficiency.
- To support data-driven decision-making through an executive dashboard that enables stakeholders to monitor business performance, compare key metrics, and evaluate strategic opportunities.
- To identify business opportunities and operational challenges by highlighting high-performing areas, underperforming segments, and profitability improvement opportunities through analytical reporting.
- To implement an end-to-end data analytics workflow involving data cleaning, exploratory data analysis, SQL querying, data modeling, DAX calculations, dashboard development, and business recommendation generation using a real-world retail dataset.

4. Dataset Description

The dataset used for this project represents historical retail sales transactions of Walmart and contains detailed information related to customer orders, product details, sales, profit, shipping, and regional performance. It was selected to perform comprehensive business analysis and develop an interactive Business Intelligence dashboard capable of supporting data-driven decision-making.

The dataset captures multiple aspects of retail operations, including customer segmentation, product hierarchy, sales transactions, profitability, shipping methods, and geographical regions. These attributes enable analysis of business performance from financial, operational, and customer perspectives.

Before analysis, the dataset underwent data quality assessment and preprocessing to identify missing values, duplicate records, inconsistencies, and outliers. The cleaned dataset served as the foundation for SQL analysis, Power BI data modeling, dashboard development, and business insight generation.

Attribute	Description
Dataset Name	Walmart Retail Sales Dataset
Total Records	8,337
Total Attributes	25
Business Domain	Retail Sales & Business Intelligence
Data Type	Structured Transactional Data
Primary Tool for Storage & Querying	MySQL
Visualization Tool	Microsoft Power BI
Programming Language	Python
Supporting Tools	Microsoft Excel, DAX

5. Data Cleaning and Preparation

Data preprocessing is a critical step in any data analytics project, as the quality of analytical insights depends directly on the quality of the underlying data. Before performing exploratory data analysis, SQL-based business analysis, and Power BI dashboard development, the Walmart Retail Sales dataset was systematically examined and prepared to ensure accuracy, consistency, and reliability.

The preprocessing phase involved importing the dataset into the Python environment, inspecting its structure, assessing overall data quality, identifying missing values, duplicate records, and statistical outliers, and applying appropriate preprocessing techniques where necessary. Each activity was performed sequentially to ensure that the dataset accurately represented the underlying business operations while preserving legitimate business transactions.

The objective of this phase was not only to clean the dataset but also to understand its overall quality and identify potential issues that could influence subsequent business analysis. By completing these preprocessing activities, the dataset was transformed into a reliable analytical foundation for SQL querying, interactive Power BI dashboard development, and business decision-making.

5.1. Dataset import

The Walmart Retail Sales dataset was provided in **Comma-Separated Values (CSV)** format and imported into the Jupyter Notebook environment using the **Pandas** library. Python was selected as the initial preprocessing environment due to its efficient data manipulation capabilities and seamless integration with analytical libraries required for subsequent preprocessing and exploratory data analysis.

Following the import, the dataset was previewed to verify that the data had been loaded successfully. Displaying the first few records provided an overview of the dataset structure, attribute names, and sample business transactions, ensuring that the imported data was complete and readable.

The dataset structure was subsequently examined to understand the number of records, available attributes, data types, and memory usage. This preliminary inspection confirmed that the dataset had been imported correctly and established the foundation

for the data quality assessment and preprocessing activities performed in the subsequent stages of the project.

At this stage, no modifications were made to the dataset. The objective of this section was solely to validate the successful import and understand the overall structure of the available business data before proceeding with detailed quality assessment.

Preview of the imported Walmart Retail Sales dataset using `data.head()`

City	Customer Age	Customer Name	Customer Segment	Discount	Number of Records	Order Date	Order ID	Order Priority	Order Quantity	...	Profit	Region	Row ID	Sales	Ship Date	Ship Mode	Shipping Cost
McKeesport	NaN	Jessica Myrick	Small Business	0.10	1	2012-01-01	28774	High	32	...	-111.800	East	4031	180.36	2012-01-02	Regular Air	4.69
Bowie	NaN	Matt Collister	Home Office	0.08	1	2012-01-01	13729	Not Specified	9	...	-342.910	East	1914	872.48	2012-01-03	Express Air	35.00
Napa	NaN	Alan Schoenberger	Corporate	0.00	1	2012-01-02	37537	Low	4	...	-193.080	West	5272	1239.06	2012-01-02	Delivery Truck	48.80
Montebello	NaN	Elizabeth Moffitt	Consumer	0.08	1	2012-01-02	44069	Critical	43	...	247.790	West	6225	614.80	2012-01-02	Regular Air	1.97
Napa	NaN	Alan Schoenberger	Corporate	0.07	1	2012-01-02	37537	Low	43	...	-1049.850	West	5273	4083.19	2012-01-04	Delivery Truck	45.00
Montebello	NaN	Elizabeth Moffitt	Consumer	0.09	1	2012-01-02	44069	Critical	16	...	26.710	West	6224	137.63	2012-01-04	Express Air	2.15
Prior Lake	NaN	David Philippe	Consumer	0.06	1	2012-01-02	9285	Critical	3	...	-11.937	Central	1279	124.81	2012-01-04	Regular Air	2.99
Napa	NaN	Alan Schoenberger	Corporate	0.05	1	2012-01-02	37537	Low	32	...	1438.490	West	5274	4902.38	2012-01-09	Regular Air	7.07

Dataset structure showing attribute names, data types, non-null values, and memory usage using `data.info()`

```
---  ------                --------------  -----
 0   City                  8399 non-null   object
 1   Customer Age          7496 non-null   float64
 2   Customer Name         8399 non-null   object
 3   Customer Segment      8399 non-null   object
 4   Discount              8399 non-null   float64
 5   Number of Records     8399 non-null   int64
 6   Order Date            8399 non-null   datetime64[ns]
 7   Order ID              8399 non-null   int64
 8   Order Priority        8399 non-null   object
 9   Order Quantity        8399 non-null   int64
 10  Product Base Margin   8336 non-null   float64
 11  Product Category      8399 non-null   object
 12  Product Container     8399 non-null   object
 13  Product Name          8399 non-null   object
 14  Product Sub-Category  8399 non-null   object
 15  Profit                8399 non-null   float64
 16  Region                8399 non-null   object
 17  Row ID                8399 non-null   int64
 18  Sales                 8399 non-null   float64
 19  Ship Date             8399 non-null   datetime64[ns]
 20  Ship Mode             8399 non-null   object
 21  Shipping Cost         8399 non-null   float64
 22  State                 8399 non-null   object
 23  Unit Price            8399 non-null   float64
 24  Zip Code              8399 non-null   int64
```

5.2. Data Quality analysis

After successfully importing and inspecting the dataset, a comprehensive data quality analysis was conducted to evaluate the reliability and consistency of the Walmart Retail Sales dataset before performing any preprocessing or business analysis. Assessing data quality at this stage ensures that the analytical results generated in later phases are based on accurate, complete, and trustworthy information.

The objective of this analysis was to identify potential data quality issues that could influence business insights or analytical outcomes. Rather than directly modifying the dataset, each quality issue was first examined to understand its nature and business significance. This approach ensured that appropriate preprocessing techniques were applied while preserving legitimate business transactions.

The data quality assessment focused on the following aspects:

- **Missing Value Analysis** – To identify incomplete records and determine the extent of missing information across all attributes.
- **Duplicate Record Analysis** – To examine the dataset for repeated records that could affect aggregation, reporting, and analytical accuracy.
- **Outlier Analysis** – To detect unusually high or low numerical values and evaluate whether they represented genuine business transactions or potential anomalies.

5.2.1. Missing value analysis

Following the initial data quality assessment, a missing value analysis was performed to identify incomplete records within the dataset. Missing values can reduce the reliability of analytical results and may introduce bias if not appropriately addressed during the preprocessing stage. Therefore, quantifying the extent of missing data was an essential step before performing any data cleaning operations.

The analysis was conducted by examining each attribute in the dataset to determine the number of missing observations. The results indicated that missing values were present in only two attributes, while all remaining variables contained complete records.

The summary of missing values identified during the assessment is presented below.

Attribute	Missing Values	Percentage (%)
Customer Age	903	10.75
Product Base Margin	63	0.75

The analysis revealed that **Customer Age** contained the highest proportion of missing values, accounting for approximately **10.75%** of the dataset, whereas **Product Base Margin** contained only **0.75%** missing observations. Since the remaining attributes were complete, the missing value issue was limited to these two variables.

Missing value count for each attribute using `data.isnull().sum()`

```
City                        0
Customer Age              903
Customer Name               0
Customer Segment            0
Discount                    0
Number of Records           0
Order Date                  0
Order ID                    0
Order Priority              0
Order Quantity              0
Product Base Margin        63
Product Category            0
Product Container           0
Product Name                0
Product Sub-Category        0
Profit                      0
Region                      0
Row ID                      0
Sales                       0
Ship Date                   0
Ship Mode                   0
Shipping Cost               0
State                       0
Unit Price                  0
Zip Code                    0
dtype: int64
```

5.2.2. Duplicate record analysis

Duplicate record analysis was performed to determine whether the dataset contained any completely repeated observations that could affect the accuracy of subsequent analysis. Duplicate records can introduce bias by inflating transaction counts, sales values, profitability measures, and other business metrics. Therefore, verifying the uniqueness of each record was an important step in the data quality assessment process.

The analysis was conducted using Python's duplicate detection functionality to identify fully identical records across all attributes in the dataset. The results indicated that the dataset **did not contain any duplicate records**, confirming that each observation represented a unique business transaction.

Since no duplicate records were identified, no duplicate removal or modification was required during the preprocessing stage.

Duplicate record verification using `data.duplicated().sum()`

```
0
```

5.2.3. Outlier analysis

Outlier analysis was conducted to identify unusually high or low numerical values that could potentially influence statistical analysis and business insights. Detecting outliers is an important step in data quality assessment because extreme observations may represent either data entry errors or genuine business events. Therefore, outliers were identified and evaluated before any preprocessing decisions were made.

The analysis was performed using the **Interquartile Range (IQR)** method, which identifies observations lying below the lower bound or above the upper bound of the expected distribution. This approach provides a robust statistical technique for detecting potential anomalies while minimizing the influence of extreme values.

To support the analysis, box plots were generated for selected numerical attributes. The visualization enabled a clear understanding of the distribution of values and highlighted observations lying outside the whisker boundaries, indicating potential outliers.

The results of the outlier assessment served as the basis for determining whether the identified observations represented legitimate business transactions or required preprocessing.

Box plot of the Customer Age attribute illustrating the distribution of values and potential outliers identified using the IQR method



5.3. Data cleaning and Feature engineering

Following the completion of the data quality assessment, appropriate preprocessing techniques were applied to prepare the dataset for exploratory data analysis, SQL-based business analysis, and Power BI dashboard development. The objective of this stage was to improve data quality while preserving the integrity of genuine business transactions.

Based on the findings from the missing value analysis, suitable imputation techniques were adopted to handle incomplete observations without unnecessarily discarding records. The **Customer Age** attribute contained a relatively high proportion of missing values and was imputed using the **median** value. Median imputation was selected because it is less sensitive to extreme values than the arithmetic mean and provides a robust estimate of the central tendency for numerical data.

Similarly, missing values in the **Product Base Margin** attribute were also treated to ensure that profitability calculations and subsequent business analysis could be performed without interruption.

In addition to data cleaning, a new analytical feature named **Delivery Days** was created by calculating the difference between the **Ship Date** and the **Order Date**. This derived attribute enabled the evaluation of delivery performance and operational efficiency during the later stages of the project.

Throughout the preprocessing stage, no duplicate records were removed because the duplicate analysis confirmed that the dataset contained no completely duplicated observations. Likewise, statistical outliers identified during the data quality assessment were retained after evaluation, as they represented legitimate business transactions rather than data entry errors. Retaining these observations ensured that the dataset continued to reflect real-world business performance.

After completing all preprocessing activities, the dataset was finalized and prepared for exploratory data analysis, SQL querying, and dashboard development.

Data Quality Issue	Action Taken
Missing Product Base Margin	Removed rows containing missing values
Missing Customer Age	Median Imputation
Duplicate Records	No Action Required
Statistical Outliers	Retained after evaluation
Delivery Days	Feature engineered using Ship Date and Order Date

Dataset after completion of data cleaning and feature engineering showing the newly created `Delivery Days` attribute

Customer Name	Customer Segment	Discount	Number of Records	Order Date	Order ID	Order Priority	Order Quantity	...	Row ID	Sales	Ship Date	Ship Mode	Shipping Cost	State	Unit Price	Zip Code	Delivery Days	Order Month
essica Myrick	Small Business	0.10	1	2012-01-01	28774	High	32	...	4031	180.36	2012-01-02	Regular Air	4.69	Pennsylvania	5.98	15131	1	1
Alan choenberger	Corporate	0.00	1	2012-01-02	37537	Low	4	...	5272	1239.06	2012-01-02	Delivery Truck	48.80	California	291.73	94559	0	1
Elizabeth Moffitt	Consumer	0.08	1	2012-01-02	44069	Critical	43	...	6225	614.80	2012-01-02	Regular Air	1.97	California	15.04	90640	0	1
Alan choenberger	Corporate	0.07	1	2012-01-02	37537	Low	43	...	5273	4083.19	2012-01-04	Delivery Truck	45.00	California	100.98	94559	2	1
Elizabeth Moffitt	Consumer	0.09	1	2012-01-02	44069	Critical	16	...	6224	137.63	2012-01-04	Express Air	2.15	California	9.11	90640	2	1

5.4. Final prepared dataset

After completing all preprocessing and feature engineering activities, the Walmart Retail Sales dataset was transformed into a clean, consistent, and analysis-ready dataset. The preprocessing phase ensured that data quality issues identified during the initial assessment were appropriately addressed while preserving the integrity of legitimate business transactions.

The final prepared dataset consisted of records with complete and reliable information for all attributes required in the subsequent stages of the project. Missing values in the **Product Base Margin** attribute were eliminated through row removal due to their negligible proportion, while missing values in **Customer Age** were treated using median imputation. In addition, the **Delivery Days** feature was successfully generated to support operational and logistics-related analysis.

The resulting dataset provided a robust foundation for exploratory data analysis, SQL-based business querying, and interactive Power BI dashboard development. By completing the preprocessing stage, the dataset was considered suitable for generating reliable business insights and supporting data-driven decision-making.

Aspect	Status
Duplicate Records	None
Missing Product Base Margin	Removed
Missing Customer Age	Median Imputed
Delivery Days	Successfully Created
Dataset Status	Ready for Analysis

Final prepared dataset after preprocessing and feature engineering

Customer Name	Customer Segment	Discount	Number of Records	Order Date	Order ID	Order Priority	Order Quantity	...	Row ID	Sales	Ship Date	Ship Mode	Shipping Cost	State	Unit Price	Zip Code	Delivery Days	Order Month
essica Myrick	Small Business	0.10	1	2012-01-01	28774	High	32	...	4031	180.36	2012-01-02	Regular Air	4.69	Pennsylvania	5.98	15131	1	1
Alan choenberger	Corporate	0.00	1	2012-01-02	37537	Low	4	...	5272	1239.06	2012-01-02	Delivery Truck	48.80	California	291.73	94559	0	1
Elizabeth Moffitt	Consumer	0.08	1	2012-01-02	44069	Critical	43	...	6225	614.80	2012-01-02	Regular Air	1.97	California	15.04	90640	0	1
Alan choenberger	Corporate	0.07	1	2012-01-02	37537	Low	43	...	5273	4083.19	2012-01-04	Delivery Truck	45.00	California	100.98	94559	2	1
Elizabeth Moffitt	Consumer	0.09	1	2012-01-02	44069	Critical	16	...	6224	137.63	2012-01-04	Express Air	2.15	California	9.11	90640	2	1

6. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics, distribution, and relationships within the Walmart Retail Sales dataset before performing SQL-based business analysis and dashboard development. This phase focused on summarizing the dataset using descriptive statistics and visualizations to uncover patterns, identify trends, and support business decision-making.

Various statistical summaries and graphical techniques were employed to examine customer demographics, sales performance, profitability, product characteristics, shipping operations, and regional business performance. The insights obtained during this stage provided the analytical foundation for the subsequent SQL queries and Power BI dashboards.

The exploratory analysis was performed using Python libraries including **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn**, enabling both numerical and visual examination of the dataset.

6.1. Descriptive Statistical Analysis

The exploratory data analysis phase commenced with the generation of descriptive statistics for the numerical attributes in the prepared dataset. Descriptive statistical analysis provides an overall understanding of the distribution, central tendency, and variability of the data before conducting detailed attribute-level analysis.

The analysis was performed using Python's `describe()` function, which summarizes key statistical measures for each numerical attribute, including:

- Count of observations
- Mean
- Standard deviation
- Minimum value
- 25th percentile (Q1)
- Median (50th percentile)
- 75th percentile (Q3)
- Maximum value

These statistics provide valuable insight into the overall characteristics of the dataset and help identify the range and spread of important business variables such as **Sales**, **Profit**, **Order Quantity**, **Shipping Cost**, **Unit Price**, **Discount**, **Customer Age**, and **Delivery Days**.

The descriptive summary also served as an initial validation of the preprocessing activities performed in Chapter 5 by confirming that the dataset contained complete numerical information suitable for further exploratory analysis.

Descriptive statistical summary of the prepared Walmart Retail Sales dataset generated using `data.describe()`

	Customer Age	Discount	Number of Records	Order Date	Order ID	Order Quantity	Product Base Margin	Profit	Row ID	Sales	Ship Date
count	8336.000000	8336.000000	8336.0	8336	8336.000000	8336.000000	8336.000000	8336.000000	8336.000000	8336.000000	8336
mean	54.376560	0.049616	1.0	2013-12-24 14:37:42.955854336	29974.350408	25.573896	0.512513	178.411589	4201.256238	1757.100276	2013-12-26 15:27:27.984644608
min	41.000000	0.000000	1.0	2012-01-01 00:00:00	3.000000	1.000000	0.350000	-14140.701600	1.000000	2.240000	2012-01-02 00:00:00
25%	47.000000	0.020000	1.0	2012-12-20 00:00:00	15033.750000	13.000000	0.380000	-83.205000	2102.750000	141.971625	2012-12-23 00:00:00
50%	53.000000	0.050000	1.0	2013-12-14 00:00:00	29858.500000	26.000000	0.520000	-1.735000	4200.500000	442.740000	2013-12-17 00:00:00
75%	60.000000	0.080000	1.0	2015-01-02 06:00:00	44609.250000	38.000000	0.590000	159.802500	6301.250000	1678.515000	2015-01-04 00:00:00
max	95.000000	0.250000	1.0	2015-12-30 00:00:00	59973.000000	50.000000	0.850000	27220.690000	8399.000000	89061.050000	2015-12-30 00:00:00
std	9.017792	0.031809	0.0	NaN	17256.903785	14.494657	0.135589	1192.010697	2424.179426	3576.433039	NaN

Shipping Cost	Unit Price	Zip Code	Delivery Days	Order Month
8336.000000	8336.000000	8336.000000	8336.000000	8336.000000
12.717320	88.716087	52857.025312	2.034549	6.523153
0.490000	0.990000	1001.000000	0.000000	1.000000
3.140000	6.480000	30338.000000	1.000000	4.000000
6.050000	20.980000	52732.000000	2.000000	7.000000
13.990000	85.990000	77583.250000	2.000000	9.250000
164.730000	6783.020000	99362.000000	92.000000	12.000000
17.200793	291.195650	28499.089156	2.307705	3.448195

6.2. Customer age distribution

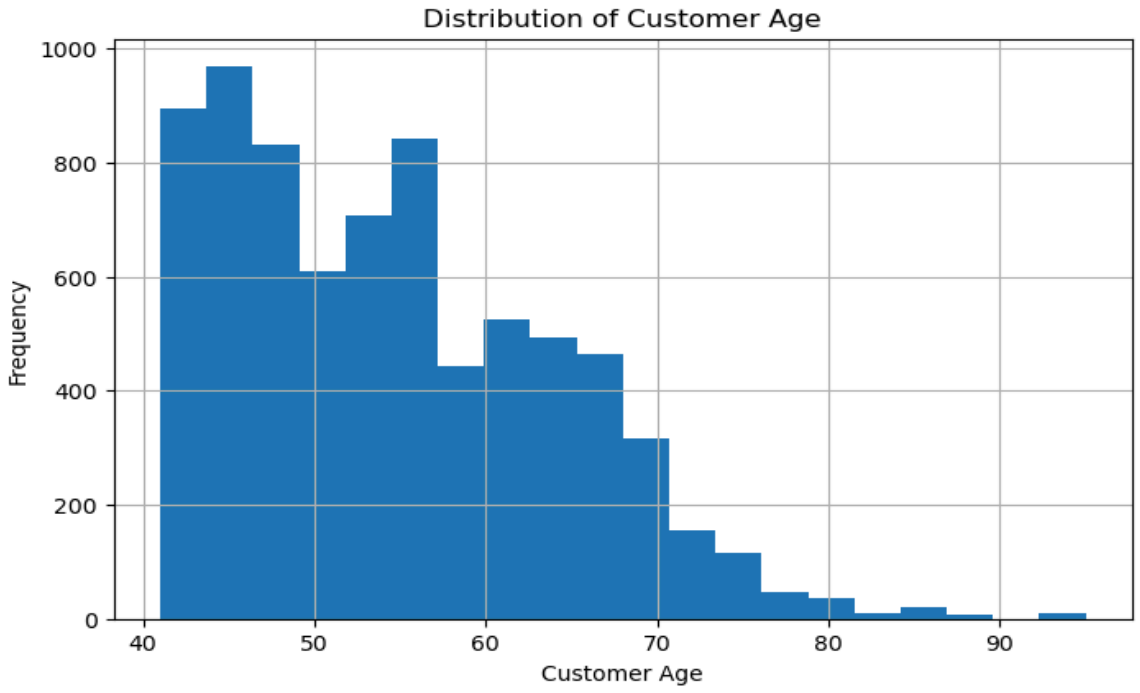
Understanding the age distribution of customers provides valuable insight into the demographic profile of the customer base and helps identify the dominant purchasing population. As part of the exploratory data analysis, the distribution of the **Customer Age** attribute was examined using a histogram.

The histogram illustrates the frequency of customers across different age groups, enabling the identification of concentration patterns, potential skewness, and the overall spread of customer ages. Examining the distribution also helps validate the effectiveness of the missing value treatment performed during the data preprocessing stage.

The visualization indicates that customer ages are distributed across a broad range, representing customers from multiple age groups. Such diversity suggests that the retail business serves a wide customer demographic rather than being concentrated within a narrow age segment.

The analysis of customer age distribution provides an important demographic perspective that can support future customer segmentation, targeted marketing strategies, and personalized business recommendations.

Histogram illustrating the distribution of customer ages in the prepared Walmart Retail Sales dataset.



6.3. Customer age box plot

To complement the histogram analysis, a box plot was generated for the **Customer Age** attribute to examine the spread of the data and identify potential outliers. Unlike a histogram, which illustrates the frequency distribution, a box plot provides a concise summary of the dataset through the median, quartiles, interquartile range (IQR), and observations lying beyond the whisker boundaries.

The visualization enabled a clearer understanding of the variability in customer ages and highlighted observations that deviated from the majority of the population. Such observations were carefully examined during the preprocessing stage to determine whether they represented genuine customer records or possible data anomalies.

The box plot indicated the presence of potential outliers within the customer age distribution. However, these observations were retained because they represented valid customer information rather than data entry errors. Preserving these values ensured that the dataset accurately reflected the diversity of the customer population.

Box plot illustrating the distribution of Customer Age and the identification of potential outliers.



6.4. Delivery Days Statistical analysis

As part of the exploratory analysis, descriptive statistics were generated for the **Delivery Days** attribute, which was created during the feature engineering phase. This derived variable represents the number of days taken between the **Order Date** and the **Ship Date**, providing a quantitative measure of delivery performance.

Statistical analysis of this attribute was performed to understand the overall delivery time distribution, identify the average delivery duration, and examine the variability in shipping performance across customer orders. The descriptive statistics include measures such as the mean, median, minimum, maximum, and quartiles, providing a comprehensive summary of delivery performance.

Understanding delivery duration is essential for evaluating operational efficiency and forms the basis for subsequent logistics-related analyses performed later in the project, including SQL-based shipping performance evaluation and Power BI operational dashboards.

Descriptive statistical summary of the Delivery Days attribute generated using `data['Delivery Days'].describe()`

```
count    8399.000000
mean        2.032028
std         2.301371
min         0.000000
25%         1.000000
50%         2.000000
75%         2.000000
max        92.000000
Name: Delivery Days, dtype: float64
```

6.5. Order Month Statistical Analysis

To understand the temporal distribution of customer orders, descriptive statistics were generated for the **Order Month** attribute. This feature was derived from the original **Order Date** during the preprocessing stage and represents the month in which each customer order was placed.

Statistical analysis of the Order Month attribute provides an overview of the distribution of transactions throughout the calendar year. By examining measures such as the

mean, median, minimum, maximum, and quartiles, it becomes possible to verify that customer orders are distributed across multiple months rather than being concentrated within a limited period.

This preliminary statistical assessment establishes a foundation for subsequent time-based business analyses, including monthly sales trends, seasonal demand patterns, and revenue comparisons performed during the SQL and Power BI phases of the project.

Descriptive statistical summary of the Order Month attribute generated using `data['Order Month'].describe()`

```
count    8399.000000
mean        6.521967
std         3.445173
min         1.000000
25%         4.000000
50%         7.000000
75%         9.000000
max        12.000000
Name: Order Month, dtype: float64
```

6.6. Numerical column analysis

Following the examination of customer demographics and derived features, the major numerical attributes of the Walmart Retail Sales dataset were analyzed individually to understand their distributions and variability. Numerical distribution analysis helps identify the spread of values, detect skewness, and understand the overall characteristics of important business metrics before performing detailed business analysis.

The analysis focused on the following key numerical attributes:

- Sales
- Profit
- Unit Price
- Shipping Cost
- Discount
- Order Quantity
- Product Base Margin
- Delivery Days

Histograms were generated for each attribute to visualize the frequency distribution of observations. These visualizations provide an understanding of transaction patterns, value concentration, and the overall range of each business metric, forming the analytical foundation for the SQL-based business analysis and Power BI dashboard development performed in the later phases of the project.

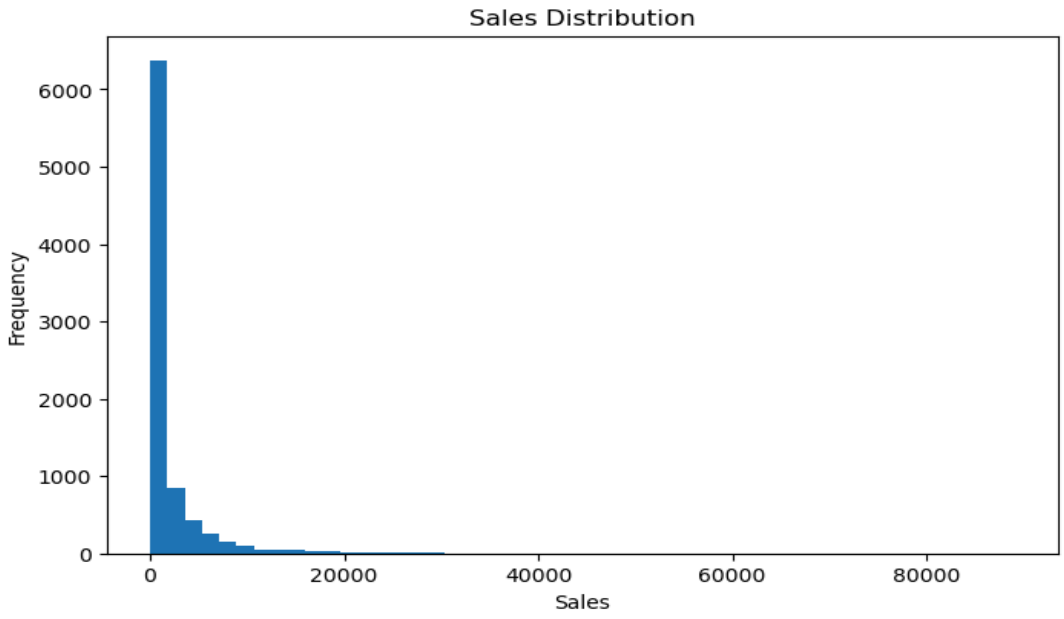
6.6.1. Sales distribution

Sales represent the primary revenue-generating metric of the retail business and provide an overall measure of transaction value across customer orders. To understand the characteristics of sales transactions, the distribution of the **Sales** attribute was examined using a histogram.

The histogram illustrates how sales values are distributed throughout the dataset by displaying the frequency of transactions across different sales ranges. Examining the distribution enables the identification of concentration patterns, variability, and the presence of exceptionally high-value transactions.

The visualization indicates that the majority of transactions are concentrated within the lower sales ranges, while comparatively fewer transactions occur at substantially higher sales values. This pattern suggests that the retail business primarily processes a large number of relatively small transactions alongside a limited number of high-value purchases.

Understanding the distribution of sales provides valuable context for subsequent profitability analysis, customer purchasing behaviour, product performance evaluation, and regional sales analysis performed during the SQL and Power BI phases of the project.



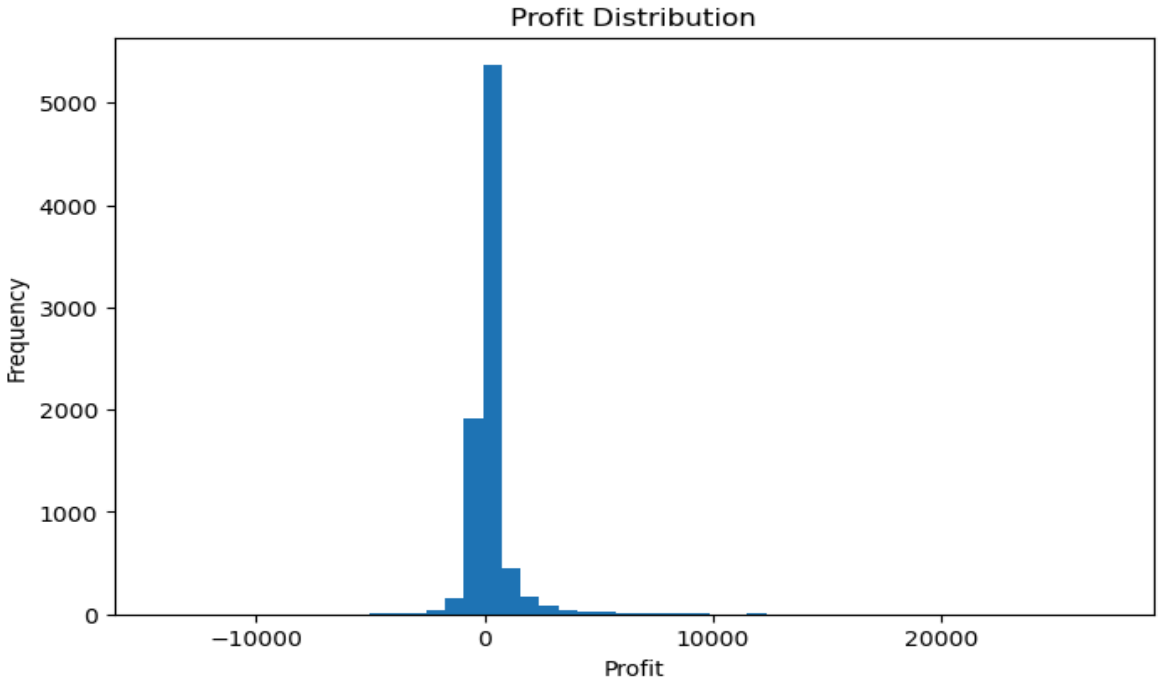
6.6.2. Profit distribution

Profit is one of the most important indicators of business performance because it reflects the financial return generated from each transaction after accounting for associated costs. To understand the profitability pattern of the retail business, the distribution of the **Profit** attribute was examined using a histogram.

The histogram illustrates the frequency of transactions across different profit ranges, enabling the identification of profitable, low-profit, and loss-making transactions. Analyzing this distribution provides valuable insight into the financial characteristics of individual sales and helps determine whether the business consistently generates profit across its transactions.

The visualization indicates that while a substantial proportion of transactions generated positive profit, a smaller number of transactions resulted in financial losses. The presence of both profitable and loss-making transactions reflects the variability typically observed in retail operations and highlights the importance of evaluating profitability alongside sales performance.

Understanding the distribution of profit establishes the foundation for subsequent analyses involving product profitability, customer profitability, regional performance, and strategic business recommendations developed through SQL analysis and Power BI dashboards.



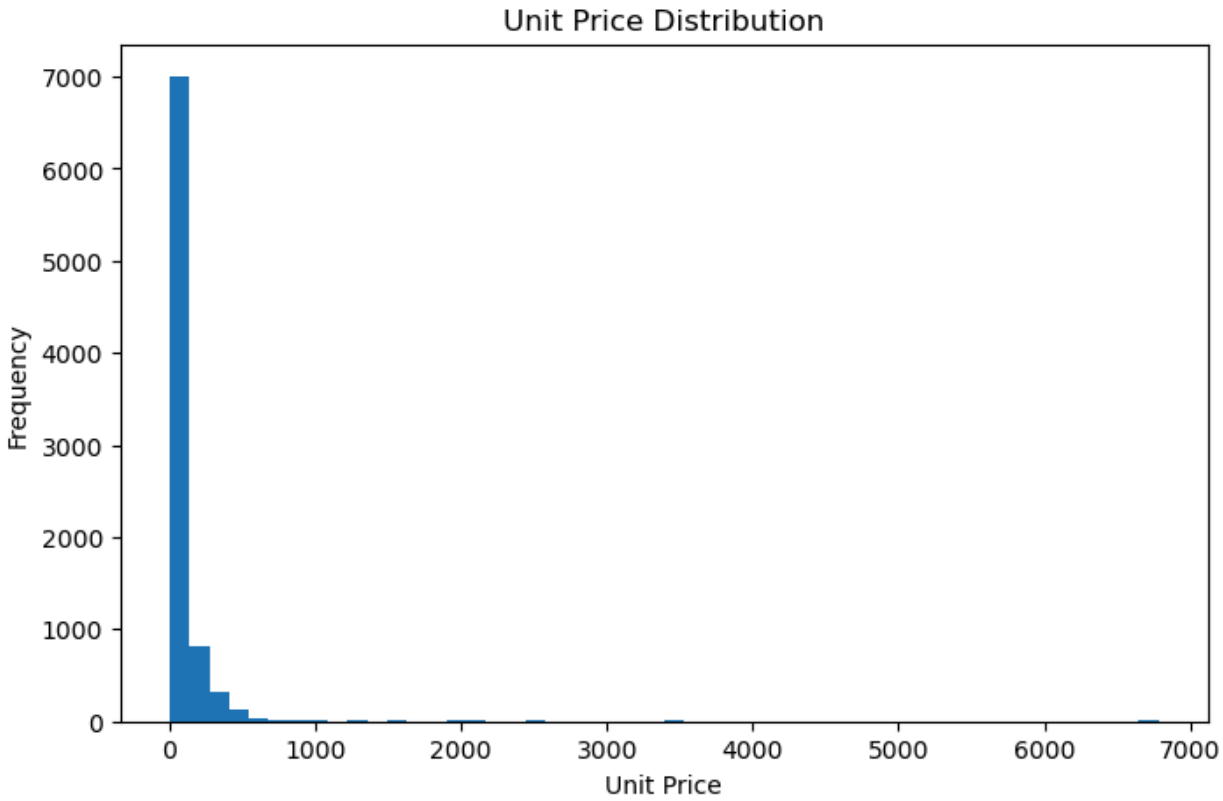
6.6.3. Unit Price distribution

The **Unit Price** attribute represents the selling price of individual products and plays a significant role in determining sales revenue and overall profitability. To understand the pricing characteristics of products within the Walmart Retail Sales dataset, the distribution of **Unit Price** was examined using a histogram.

The histogram illustrates the frequency of products across different price ranges, providing insight into the overall pricing structure of the retail business. Analyzing this distribution helps identify whether the business primarily deals with low-priced products, premium-priced products, or a combination of both.

The visualization indicates that product prices are distributed across a wide range, reflecting the diversity of products offered by the retail business. While a large proportion of products fall within the lower price ranges, comparatively fewer products are available at higher price levels. This pattern is consistent with retail environments that offer products across multiple pricing categories to serve different customer segments.

Understanding the distribution of unit prices provides an important foundation for analyzing product performance, sales trends, profitability, and pricing strategies in the subsequent stages of the project.



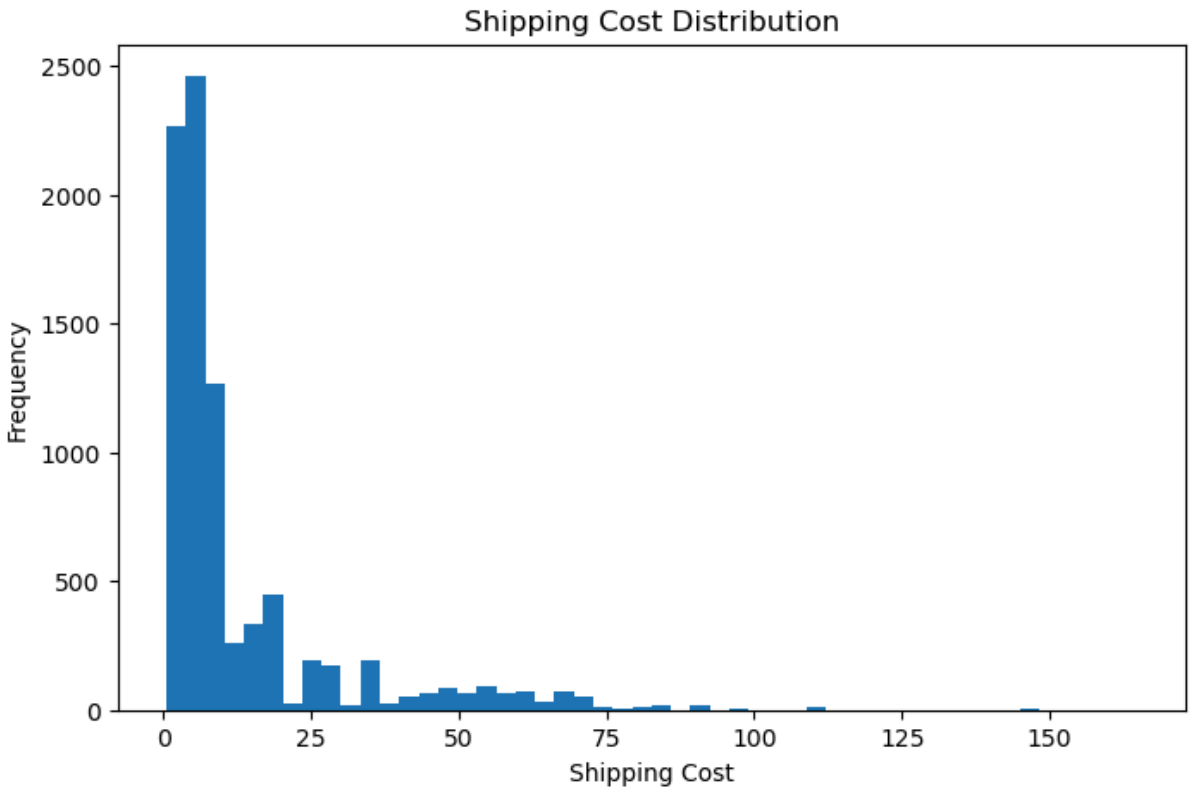
6.6.4 Shipping Cost Distribution

Shipping Cost is an important operational metric that directly influences logistics expenditure and overall business profitability. To understand the distribution of shipping expenses across customer orders, the **Shipping Cost** attribute was analyzed using a histogram.

The histogram illustrates the frequency of transactions across different shipping cost ranges, enabling the identification of common shipping cost patterns and the overall variability in logistics expenses. Examining this distribution provides valuable insight into the operational characteristics of the retail business and helps determine whether shipping costs are concentrated within a narrow range or vary significantly across transactions.

The visualization indicates that shipping costs are distributed across multiple cost ranges, with a greater concentration of transactions occurring at lower shipping costs and comparatively fewer transactions incurring higher logistics expenses. This pattern reflects the natural variation in shipping costs resulting from differences in product characteristics, order quantities, shipping methods, and delivery destinations.

Analyzing shipping cost distribution establishes an important foundation for evaluating logistics efficiency, operational expenditure, and their relationship with sales and profitability in the subsequent phases of the project.



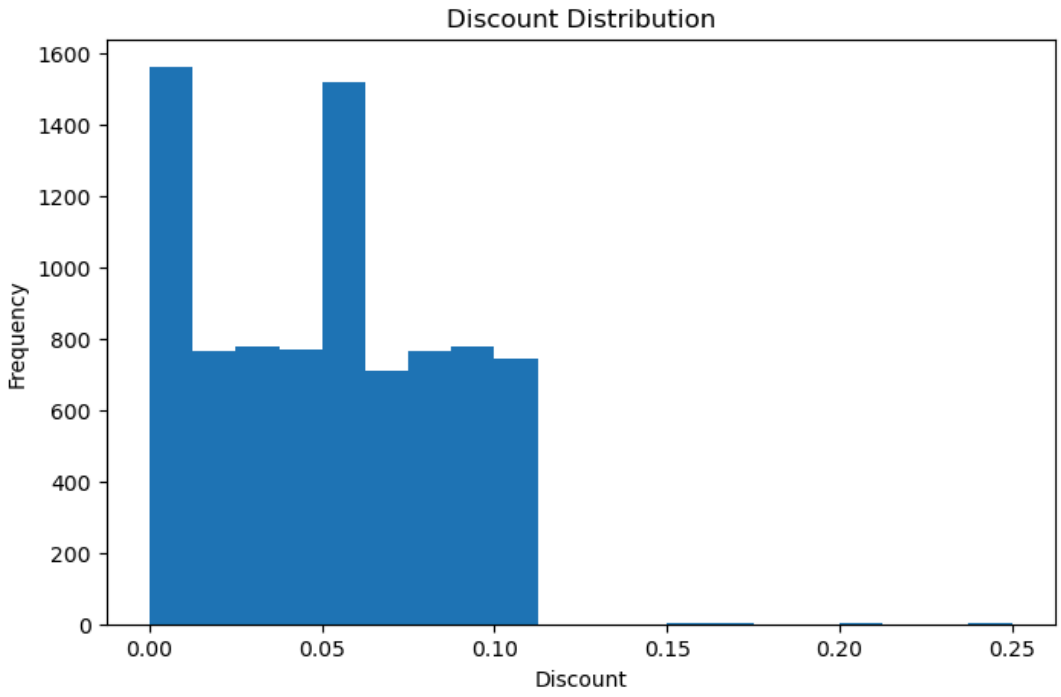
6.6.5 Discount Distribution

Discount is an important pricing strategy that influences customer purchasing behaviour, sales volume, and overall profitability. To understand how discounts were applied across customer transactions, the distribution of the **Discount** attribute was analyzed using a histogram.

The histogram illustrates the frequency of transactions across different discount values, providing an overview of the organization's discounting practices. Examining the distribution enables the identification of commonly applied discount levels and the variability in promotional pricing across customer orders.

The visualization indicates that a large proportion of transactions were completed with relatively low discount values, while higher discount percentages were applied less frequently. This distribution suggests that the retail business primarily relies on moderate discounting strategies while reserving larger discounts for a comparatively smaller number of transactions.

Analyzing the distribution of discounts provides valuable insight into pricing behaviour and establishes the foundation for evaluating the relationship between discounts, sales performance, and profitability during the subsequent SQL analysis and Power BI dashboard development.



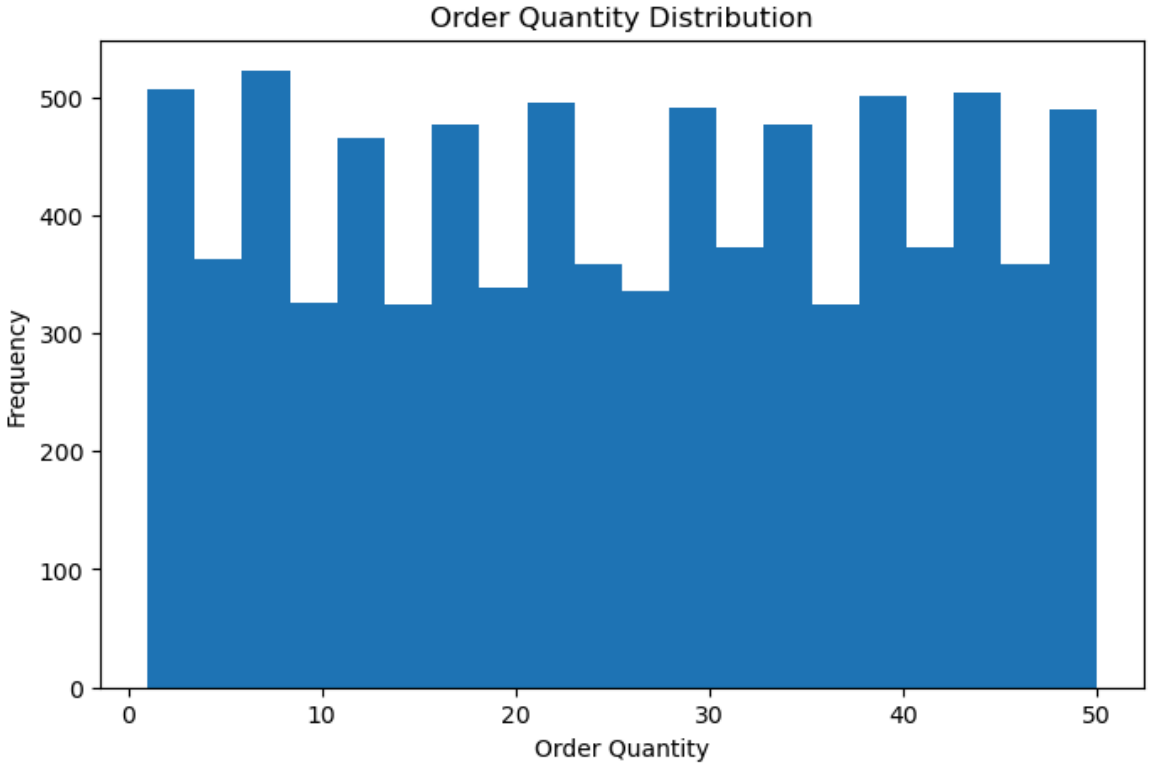
6.6.6 Order Quantity Distribution

Order Quantity represents the number of product units purchased in each transaction and provides valuable insight into customer purchasing behaviour and transaction size. To understand the distribution of quantities ordered across the dataset, the **Order Quantity** attribute was analyzed using a histogram.

The histogram illustrates the frequency of transactions for different order quantities, enabling an assessment of typical purchasing patterns and the variability in customer order sizes. Examining this distribution helps determine whether customers generally purchase single items or multiple units within a transaction.

The visualization indicates that a majority of transactions involve relatively small order quantities, while larger quantity purchases occur less frequently. This distribution reflects the typical purchasing behaviour observed in retail environments, where individual or low-volume purchases are more common than bulk orders.

Analyzing the distribution of order quantities provides an important foundation for understanding customer purchasing patterns and supports subsequent analyses related to inventory management, sales performance, and demand forecasting.



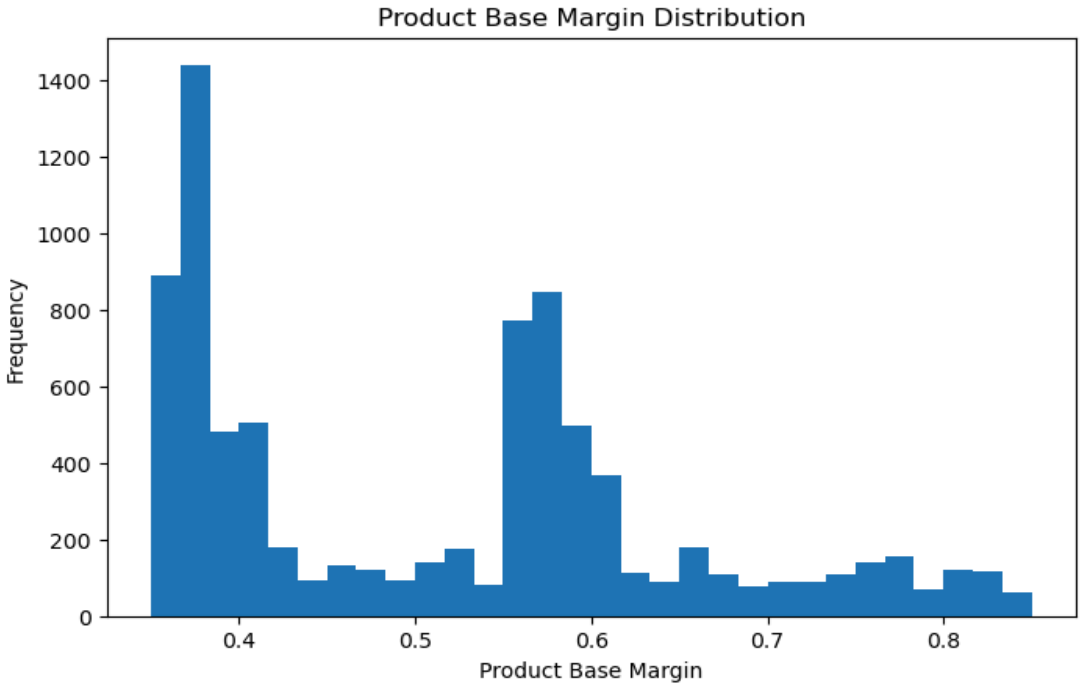
6.6.7 Product Base Margin Distribution

Product Base Margin is an important profitability-related attribute that represents the margin associated with individual products before considering additional operational expenses. Understanding its distribution provides insight into the overall profitability characteristics of the products included in the Walmart Retail Sales dataset.

To examine this attribute, a histogram was generated to visualize the frequency distribution of **Product Base Margin** values across all transactions. The visualization enables an assessment of how product margins vary throughout the dataset and whether most products operate within similar profitability ranges.

The histogram indicates that Product Base Margin values are distributed across multiple margin levels, reflecting variation in profitability among different products. While a considerable proportion of products are concentrated within moderate margin ranges, relatively fewer products exhibit extremely low or high base margins. Such variation is expected in a retail environment that offers a diverse portfolio of products with different pricing strategies and cost structures.

The analysis of Product Base Margin establishes a useful foundation for evaluating product profitability, supporting later SQL-based business analysis and Power BI dashboards focused on revenue and profit performance.



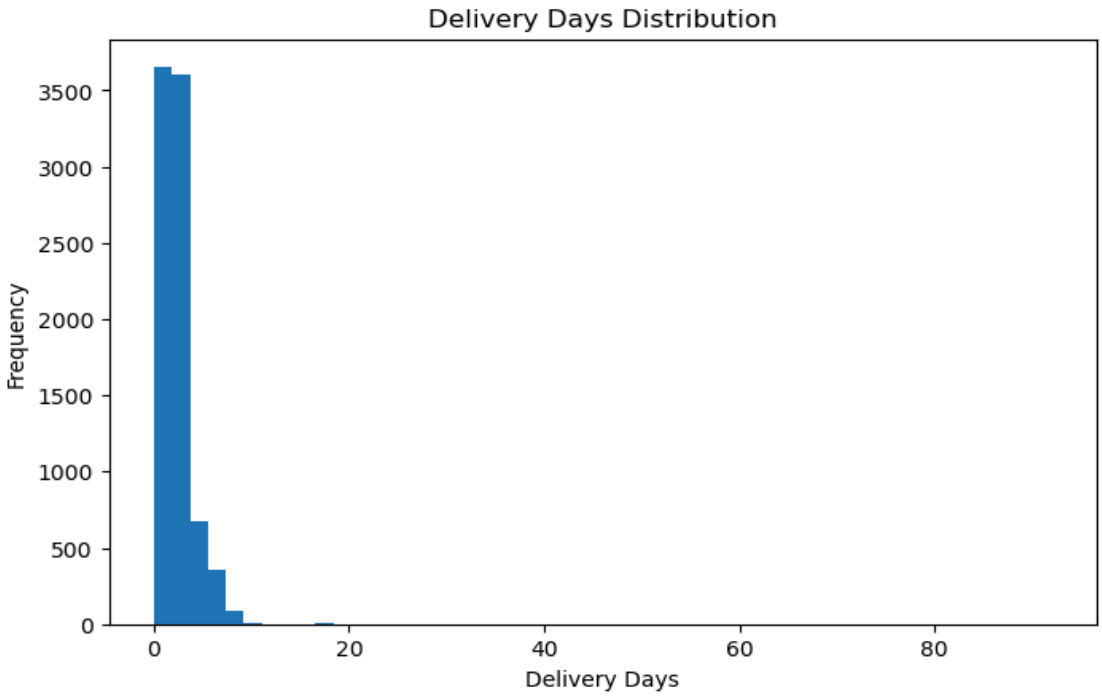
6.6.8 Delivery Days Distribution

Delivery performance is a key operational metric that reflects the efficiency of order fulfillment and logistics management. Following the creation of the **Delivery Days** feature during the preprocessing stage, its distribution was examined using a histogram to understand the variation in delivery duration across customer orders.

The histogram illustrates the frequency of orders corresponding to different delivery durations, providing insight into the consistency of the organization's shipping process. Analyzing this distribution helps determine whether deliveries are concentrated within a typical timeframe or exhibit significant variability across transactions.

The visualization indicates that a majority of customer orders were delivered within a relatively limited range of delivery days, while comparatively fewer transactions experienced longer delivery durations. This pattern suggests that the organization maintains a generally consistent delivery process, with only a small proportion of orders requiring extended shipping time.

Understanding the distribution of delivery duration provides valuable operational insight and serves as a foundation for subsequent analyses involving shipping efficiency, logistics performance, and service quality.



7. SQL Business analysis

The prepared Walmart Retail Sales dataset was imported into **MySQL** to perform structured business analysis using Structured Query Language (SQL). SQL enables efficient retrieval, aggregation, filtering, and analysis of large volumes of structured data, making it an essential tool for extracting meaningful business insights from transactional datasets.

The primary objective of this phase was to investigate various business scenarios by querying the prepared dataset. The analyses performed in this chapter focus on evaluating data integrity, sales performance, profitability, customer purchasing behaviour, product performance, regional performance, logistics efficiency, and operational metrics. Each analysis was designed to answer specific business questions using SQL techniques such as aggregate functions, filtering, grouping, sorting, joins, Common Table Expressions (CTEs), window functions, ranking functions, and conditional expressions.

Unlike the exploratory data analysis presented in the previous chapter, which focused on understanding the characteristics of the dataset through statistical summaries and visualizations, the SQL analysis emphasizes answering business-oriented questions through structured database queries. The results obtained from these analyses provide actionable insights that support strategic decision-making and serve as the analytical foundation for the interactive Power BI dashboards developed in the subsequent chapter.

The chapter begins with database validation to ensure the integrity and consistency of the imported dataset before proceeding with detailed business analyses across multiple operational and financial dimensions.

7.1. Database validation

Before performing business-oriented analysis, the prepared Walmart Retail Sales dataset was validated within the MySQL environment to ensure that the imported data was complete, consistent, and suitable for analytical processing. Database validation is an essential step in the analytical workflow because it verifies the integrity of the dataset after import and ensures that subsequent SQL analyses are based on reliable information.

The validation process began by confirming the successful creation and population of the `walmart_sales` table after loading data from the staging table. The total number of records was verified to ensure that all cleaned observations were successfully

imported into the analytical database. The table structure was then examined using the `DESCRIBE` statement to confirm that each attribute had been assigned the appropriate data type required for subsequent analytical operations.

To maintain data integrity, duplicate records were identified using the unique `row_id` field to verify that no duplicate transactions existed within the dataset. A comprehensive missing value validation was also performed across all business attributes, including customer information, product information, sales metrics, shipping details, and operational variables. These validation queries confirmed that the preprocessing activities completed during the data preparation phase had successfully eliminated missing values from the analytical dataset.

Additional business validation checks were conducted to identify loss-making transactions, detect invalid negative sales values, and verify the chronological range of order dates contained within the database. These checks ensured that the dataset accurately represented valid business transactions and that no inconsistencies remained before commencing analytical querying.

The successful completion of the database validation process confirmed that the Walmart Retail Sales database was structurally sound, free from duplicate records, consistent with the cleaned dataset, and suitable for advanced SQL-based business analysis presented in the subsequent sections of this chapter.

7.2. Key Business Performance Indicators (KPIs)

Key Performance Indicators (KPIs) were calculated to obtain a comprehensive overview of the overall business performance before conducting detailed analytical investigations. These indicators provide management with a concise summary of the organization's operational and financial performance by measuring overall sales, profitability, order volume, customer purchasing behaviour, and business efficiency.

Using SQL aggregate functions such as **SUM()**, **COUNT()**, and arithmetic calculations, five core performance indicators were derived from the Walmart Retail Sales database. The calculated metrics include **Total Sales**, **Total Profit**, **Total Orders**, **Average Order Value**, and **Profit Margin Percentage**. These KPIs establish the baseline for evaluating subsequent analyses related to products, customers, regions, and operational performance. The SQL queries utilize aggregate operations over the complete transaction dataset to summarize business performance at the organizational level.

The KPI calculations indicate that the organization generated **$14.65 million** in total sales while achieving an overall profit of **$1.49 million** from **5,474 customer orders**.

The analysis also provides the Average Order Value, which reflects the average revenue generated per customer order, and the overall Profit Margin Percentage, representing the proportion of revenue retained as profit. Together, these indicators provide a high-level assessment of business health and serve as reference metrics throughout the remaining SQL analyses and the Power BI dashboard developed in the subsequent chapter.

7.3. Monthly Business Performance analysis

Monthly business performance analysis was conducted to evaluate how sales and profit varied over time and to identify patterns in business performance across different months. Time-based analysis enables organizations to monitor revenue growth, recognize seasonal demand fluctuations, assess profitability trends, and identify periods of exceptional or declining performance. Such insights support strategic planning, inventory management, budgeting, and sales forecasting.

The SQL queries executed in this section aggregated transactional data by the **Order Month** field to calculate monthly sales, monthly sales contribution percentages, and monthly profit. Aggregate functions such as **SUM()**, along with **GROUP BY** and **ORDER BY**, were utilized to summarize business performance at the monthly level. Additionally, percentage calculations were performed to determine each month's contribution to the organization's overall revenue, providing a clearer understanding of the relative importance of individual months in total business performance.

The monthly sales trend analysis provides a chronological view of revenue generation throughout the business period, while the monthly sales contribution analysis identifies months that contributed the highest proportion of total sales. Similarly, the monthly profit analysis evaluates profitability across different months, allowing management to distinguish between periods of strong revenue generation and periods of higher financial returns. Together, these analyses offer a comprehensive understanding of business performance over time and establish the foundation for identifying seasonal business patterns.

The results obtained from these SQL analyses directly support the **Monthly Sales Trend** visualization presented in the Power BI dashboard, ensuring consistency between the database analysis and the interactive business intelligence reporting developed in the subsequent phase of the project.

7.4. Product category analysis

Product category analysis was performed to evaluate the contribution of each major product category to the overall sales and profitability of the retail business. Analyzing performance at the category level enables organizations to identify high-performing product segments, assess category-wise profitability, and support strategic decisions related to inventory planning, pricing strategies, marketing investments, and resource allocation.

The SQL queries in this section aggregated transactional data by **Product Category** to calculate total sales, total profit, and profit margin for each category. Aggregate functions such as **SUM()**, together with **GROUP BY** and percentage calculations, were utilized to compare the financial performance of the three major product categories: **Technology**, **Furniture**, and **Office Supplies**. The profit margin analysis further evaluated the efficiency with which each category converted sales revenue into profit, providing a more comprehensive assessment than sales performance alone.

The analysis revealed clear differences in category performance. **Technology** generated the highest overall sales and profitability while also recording the strongest profit margin, making it the organization's best-performing product category. **Office Supplies** demonstrated stable sales performance with healthy profitability, whereas **Furniture**, despite contributing substantially to total sales, generated comparatively lower profit margins, indicating opportunities for cost optimization and pricing improvements. These findings highlight that revenue generation alone does not necessarily translate into higher profitability and emphasize the importance of evaluating both sales and financial returns simultaneously. The corresponding Power BI dashboard also reflects these trends, showing Technology as the leading category in terms of sales, profit, and profit margin.

Overall, the product category analysis provides valuable insights into category-wise business performance and supports strategic decisions regarding product portfolio management, investment prioritization, and long-term profitability improvement.

7.5. Regional analysis

Regional analysis was performed to evaluate the geographical distribution of business performance across the four operating regions: **Central, West, East, and South**. Understanding regional performance is essential for identifying markets that contribute the highest revenue and profit, evaluating geographical disparities, and supporting strategic decisions related to market expansion, resource allocation, and regional business planning.

The SQL queries executed in this section aggregated transactional data using the **Region** attribute to calculate total sales and total profit generated by each region. Aggregate functions such as **SUM()**, combined with **GROUP BY** and **ORDER BY**, were used to rank regions according to their financial performance and facilitate meaningful comparisons between geographical markets.

The regional sales analysis identified the **Central region** as the highest revenue-generating market, followed by the **West**, **East**, and **South** regions. However, revenue generation alone did not fully reflect financial performance. The regional profit analysis demonstrated differences in profitability across these markets, indicating that higher sales do not always correspond to higher financial returns. This comparison provides valuable insight into the efficiency of business operations across different geographical locations and helps identify regions that may require targeted operational or strategic improvements.

The findings from the SQL analysis are consistent with the regional performance visualizations presented in the Power BI dashboard, where regional sales and profit metrics are displayed to support comparative business evaluation. Together, these analyses provide management with a comprehensive understanding of regional business performance and establish a foundation for data-driven decision-making related to geographical expansion, sales strategy, and profitability optimization.

7.6. Customer analysis

Customer analysis was conducted to evaluate the purchasing behaviour and financial contribution of different customer segments within the Walmart Retail Sales dataset. Understanding customer segment performance enables organizations to identify their most valuable customer groups, assess purchasing patterns, and develop targeted marketing and customer retention strategies that maximize long-term business value.

The SQL analysis grouped transactional records by **Customer Segment** and calculated the number of distinct orders, total sales, and total profit generated by each segment. The analysis was performed using aggregate functions such as **COUNT(DISTINCT)** and **SUM()**, together with **GROUP BY** and **ORDER BY**, to compare the contribution of each customer segment to the organization's overall business performance.

The results indicate that the **Corporate** segment recorded the highest number of orders and generated the largest share of total sales, making it the organization's most significant revenue-generating customer segment. The **Home Office**, **Small Business**, and **Consumer** segments also contributed substantially to overall business performance, although with comparatively lower sales volumes. Profit analysis across

customer segments further demonstrates that the Corporate segment generated the highest profit, while the remaining segments maintained relatively balanced profitability despite differences in order volume and revenue contribution. These findings provide valuable insights for customer segmentation, targeted promotional campaigns, and customer relationship management initiatives. The corresponding Power BI dashboard reinforces these observations through visualizations of **Orders by Customer Segment**, **Sales by Customer Segment**, and **Profit by Customer Segment**.

7.7. Product analysis

Product analysis was performed to identify the products that contributed most significantly to the organization's sales and profitability while also ensuring the integrity of product-related information within the analytical database. Evaluating product-level performance enables organizations to recognize high-performing products, identify revenue and profit drivers, and support strategic decisions related to inventory management, product portfolio optimization, and pricing strategies.

The SQL analysis began by identifying the **Top 10 Products by Sales**, where transactional data was aggregated using the **Product Name** attribute to determine the products generating the highest revenue. During this analysis, validation queries were also executed to identify missing or empty product names. Records containing blank product names were standardized by replacing empty values with **"Unknown Product"**, ensuring consistency in product-level reporting and preventing incomplete records from affecting analytical results. Following this data validation step, the **Top 10 Most Profitable Products** were identified by aggregating total profit for each product and ranking them in descending order. These queries employed aggregate functions, filtering conditions, and ordering operations to evaluate product performance from both revenue and profitability perspectives.

The analysis revealed that **Unknown Product** generated the highest sales, indicating that a small number of transactions contained missing product descriptions that were standardized during data validation. Among the identified products, **Global Troy™ Executive Leather Chair**, **Polycom ViewStation**, and **Canon PC940 Copier** emerged as some of the highest revenue-generating products. Profitability analysis demonstrated that several of these products also ranked among the organization's most profitable offerings, confirming their strategic importance within the product portfolio. The corresponding Power BI dashboard presents these findings through the **Top 10 Products by Sales** and **Top 10 Most Profitable Products** visualizations, providing business users with an interactive view of product-level performance.

7.8. Logistics and Operational analysis

Logistics and operational analysis was conducted to evaluate the efficiency of the organization's order fulfillment process and to verify the accuracy of operational attributes used throughout the analytical database. Efficient logistics operations play a crucial role in customer satisfaction, transportation cost management, and overall supply chain performance. This section focuses on analysing delivery performance, shipping costs, shipping methods, and validating engineered operational features generated during the data preparation phase.

The SQL queries calculated the **average delivery time** and **average shipping cost** across all transactions to establish baseline operational performance. Further analysis grouped orders by **Ship Mode** to compare the average delivery duration and order volume handled by each shipping method. These queries utilized aggregate functions such as **AVG()**, **COUNT()**, and **GROUP BY** to evaluate the operational characteristics of different transportation methods.

In addition to operational metrics, validation queries were executed to verify the correctness of engineered features created during preprocessing. The first validation compared the stored **Delivery_Days** attribute against the calculated difference between **Ship_Date** and **Order_Date** using the `DATEDIFF()` function. The second validation confirmed that the generated **Order_Month** attribute matched the month extracted directly from **Order_Date** using the `MONTH()` function. These checks ensured that the derived variables used in both SQL analysis and Power BI visualizations were accurately generated and free from inconsistencies.

The analysis indicates that the average delivery time remained approximately **two days** across all shipping modes, demonstrating a consistent delivery process irrespective of transportation method. However, shipping costs varied considerably, with **Delivery Truck** representing the highest average shipping cost, while **Regular Air** and **Express Air** maintained comparatively lower average shipping expenses. The validation queries returned no inconsistencies, confirming that the engineered operational attributes accurately represented the underlying transactional data. These findings are consistent with the logistics visualizations presented in the Power BI dashboard, including **Sales by Ship Mode**, **Profit by Ship Mode**, **Average Delivery Days by Ship Mode**, and **Average Shipping Cost by Ship Mode**.

7.9. Advanced SQL analysis

To extend the business analysis beyond descriptive reporting, advanced SQL techniques were employed to extract deeper operational and strategic insights from the

Walmart Retail Sales database. This section utilizes **Common Table Expressions (CTEs)**, **window functions**, **ranking functions**, **conditional logic (CASE)**, and **cumulative aggregations** to answer complex business questions that cannot be addressed through basic aggregation alone. These analyses provide decision-makers with a more comprehensive understanding of customer behaviour, product performance, regional contributions, and operational efficiency.

The advanced analyses include **Product Profit Ranking**, which ranks products according to their overall profitability using the `RANK()` window function, enabling the identification of the organization's highest profit-generating products. A **Repeat Customer Analysis** identifies customers who placed multiple orders, providing insight into customer loyalty and repeat purchasing behaviour. The **State-wise Sales Ranking** evaluates geographical performance by ranking states according to total sales revenue, while the **Running Revenue Total** calculates cumulative monthly sales using window functions to illustrate business growth over time.

Operational performance was further evaluated through a **Delivery Performance Classification**, where delivery durations were categorized into *Fast Delivery*, *Normal Delivery*, and *Delayed Delivery* using a `CASE` expression. Additional analyses identified **High Revenue but Low Profit Products**, highlighting products that generate substantial revenue while producing negative overall profit, thereby indicating potential pricing or cost management issues. A **Discount Impact Analysis** examined how different discount levels influenced both sales and profitability, supporting pricing strategy evaluation.

To support strategic decision-making, profitability comparisons were extended through **Customer Segment Profitability Ranking**, which calculated profit margins for each customer segment. A **Region Contribution Percentage** analysis measured the proportion of total organizational sales generated by each region, while the **Best Selling Product in Each Category** analysis identified the highest revenue-generating product within every product category using `ROW_NUMBER()` partitioning. Collectively, these advanced SQL analyses transform transactional data into actionable business intelligence by revealing hidden performance patterns, ranking business entities, validating operational efficiency, and supporting strategic recommendations presented in the Power BI dashboard.

8. Power BI Dashboard development

Following data preprocessing in Python and business analysis using SQL, the cleaned dataset was imported into Microsoft Power BI to transform analytical results into

interactive business intelligence reports. The dashboard was designed to present organizational performance through intuitive visualizations, allowing users to analyse sales, profitability, customer behaviour, product performance, regional distribution, and operational metrics within a single reporting environment.

The dashboard development process involved importing the prepared dataset, establishing the required data model, creating calculated measures using DAX, designing interactive visualizations, and implementing filters to enhance analytical flexibility. Each report page was developed to represent a specific business domain while maintaining consistency in design, navigation, and user experience. The dashboard enables stakeholders to explore business performance at multiple levels without directly interacting with the underlying transactional database.

8.1. Data Model Verification & Preparation

The first stage of the Power BI development process focused on verifying the imported dataset and preparing the analytical model before dashboard creation. The objective of this phase was to ensure that the cleaned dataset generated during the Python preprocessing stage was accurately imported into Power BI and was suitable for developing reliable business intelligence reports. Performing this verification before visualization development minimized the possibility of inconsistencies in business metrics and ensured that subsequent dashboard outputs accurately reflected the underlying data.

After importing the cleaned dataset into Power BI, the imported fields were systematically reviewed to verify that all records, attributes, and values matched the cleaned dataset produced during the preprocessing phase. Each column was inspected to confirm that the appropriate data type had been assigned. Numerical fields, categorical attributes, and date fields were validated to ensure compatibility with report filters, aggregations, and visualizations. Since the project utilized a single cleaned analytical dataset, no additional table relationships were required, thereby maintaining a simple and efficient analytical model.

The imported data was further validated by comparing sample records and aggregated business values with the outputs previously obtained during the SQL analysis. This verification ensured consistency between Python preprocessing, SQL business analysis, and the Power BI reporting environment. Only after confirming that the imported data accurately represented the analytical dataset did the project proceed to the dashboard planning and visualization development stage.

8.2. Dashboard Planning

After verifying the imported dataset, the next stage focused on planning the overall dashboard structure before creating individual visualizations. Rather than directly developing charts, the dashboard was first designed from a business perspective to ensure that the final report would answer the key analytical questions identified during the SQL analysis. This planning phase established the overall layout, determined the sequence of report pages, and identified the business metrics that would be presented throughout the dashboard.

The dashboard was organized into multiple report pages, with each page dedicated to a specific business domain. This modular design improved report readability and allowed users to analyse different aspects of organizational performance independently while maintaining consistency across the entire reporting solution. The dashboard pages were planned to present executive-level business performance, sales analysis, customer behaviour, shipping and operational performance, and overall business insights obtained during the earlier phases of the project.

Before visualization development, appropriate visual types were selected based on the nature of the business question being addressed. KPI cards were planned for displaying high-level performance indicators, bar and column charts were selected for comparative analysis, line charts were reserved for trend analysis, pie and donut charts were considered for proportional comparisons where appropriate, and geographical visualizations were planned for regional business analysis. This planning process ensured that each visualization effectively communicated the intended business insight while maintaining consistency in design and interpretation.

A common dashboard theme, standardized colour palette, and consistent formatting guidelines were also established during this phase to provide a uniform appearance across all report pages. Titles, legends, labels, and visual alignment were planned before implementation to improve readability and provide a consistent user experience throughout the dashboard. Only after completing this planning process did the project proceed to the development of the individual dashboard components.

8.3. Executive Dashboard Development

Following the completion of dashboard planning, the Executive Dashboard was developed as the primary reporting interface of the Power BI solution. The objective of this dashboard was to provide stakeholders with a consolidated overview of the organization's overall business performance through high-level business indicators and

summary visualizations. This page was designed to enable users to quickly evaluate organizational performance without navigating through multiple analytical reports.

The development process began by identifying the key performance indicators that best represented the overall business performance. Visual elements were then arranged to maintain a logical information flow, ensuring that the most important business metrics were immediately visible upon opening the dashboard. Appropriate visualizations were selected to present organizational performance in a concise and easily interpretable format while maintaining consistency with the business analyses previously performed using SQL.

During implementation, emphasis was placed on maintaining a clean dashboard layout by ensuring consistent visual spacing, standardized formatting, uniform colour selection, and proper alignment of all report components. Visual titles, labels, legends, and formatting properties were configured to improve readability and provide a consistent reporting experience across the dashboard. The placement of visuals was carefully planned to reduce clutter while allowing users to identify important business trends with minimal effort.

Before finalizing the dashboard, the displayed business metrics were validated against the SQL query results to ensure consistency between the analytical database and the Power BI reporting environment. This verification confirmed that the Executive Dashboard accurately represented the overall business performance and served as the foundation for the detailed analytical dashboards developed in the subsequent stages of the project.

8.4. Sales Dashboard Development

The Sales Dashboard was developed to provide a comprehensive analysis of the organization's sales performance across multiple business dimensions. The objective of this dashboard was to transform the sales-related findings obtained during the SQL analysis into interactive visualizations that enable users to monitor revenue generation, identify sales trends, and compare business performance across regions, product categories, customer segments, and shipping methods.

The dashboard development process began by identifying the sales-related business questions that required visualization. Appropriate charts were then selected based on the analytical objective of each business metric. Comparative visualizations were developed to evaluate sales across different business categories, while trend-based visualizations were implemented to analyse changes in sales performance over time.

The arrangement of visuals was carefully planned to present business information in a logical sequence, allowing users to interpret sales performance efficiently.

During dashboard development, visual formatting was standardized to maintain consistency throughout the report. Titles, axis labels, legends, data labels, colour schemes, and alignment properties were configured to improve readability and ensure that each visualization clearly communicated the intended business insight. Interactive filtering capabilities were incorporated to allow users to analyse sales performance across different business dimensions without modifying the underlying dataset.

Upon completion of the dashboard, the sales values displayed in each visualization were verified against the corresponding SQL query results to ensure consistency between the analytical database and the reporting environment. This validation confirmed that the Sales Dashboard accurately represented the organization's sales performance and provided a reliable foundation for interactive business analysis.

8.5. Customer Dashboard Development

The Customer Dashboard was developed to analyse customer behaviour and evaluate the contribution of different customer segments to the organization's overall business performance. The objective of this dashboard was to convert the customer-related findings obtained during the SQL analysis into interactive visualizations that enable users to identify purchasing patterns, compare customer segment performance, and assess the overall distribution of revenue and profit across the customer base.

The development process began by identifying the key customer performance indicators that required visualization. Visualizations were then designed to compare customer segments based on sales, profit, and order volume while presenting customer-related business information in a structured and easily interpretable manner. Appropriate chart types were selected to facilitate comparison between customer groups and to highlight the contribution of each segment towards the organization's overall performance.

During implementation, consistent dashboard formatting practices were maintained to ensure visual uniformity across all report pages. Titles, legends, axis labels, colour schemes, and visual alignment were standardized to improve readability and provide a seamless analytical experience. Interactive filtering capabilities were incorporated to enable users to analyse customer performance under different business conditions without modifying the underlying dataset.

Following the completion of the Customer Dashboard, all displayed values were validated against the corresponding SQL query outputs to ensure analytical consistency. This verification confirmed that the dashboard accurately represented customer-related business metrics and provided reliable insights for evaluating customer behaviour and segment performance within the organization.

8.6 Shipping & Operations Dashboard Development

The Shipping and Operations Dashboard was developed to analyse the operational efficiency of the organization's order fulfilment process. The objective of this dashboard was to visualize logistics-related business metrics identified during the SQL analysis, enabling users to evaluate shipping performance, delivery efficiency, shipping costs, and operational trends through an interactive reporting environment.

The development process focused on selecting visualizations capable of effectively representing operational performance across different shipping methods and business dimensions. Appropriate charts were created to compare shipping modes, evaluate average delivery duration, analyse shipping costs, and examine their impact on overall business performance. The arrangement of the visuals followed a structured layout to ensure that users could interpret operational performance efficiently while maintaining consistency with the overall dashboard design.

During implementation, all operational visualizations were formatted using standardized design principles established during the dashboard planning phase. Consistent titles, axis formatting, colour schemes, legends, and data labels were applied across all visuals to maintain a uniform reporting experience. Interactive filtering and cross-filtering capabilities were incorporated to allow users to examine logistics performance for specific customer segments, regions, product categories, and shipping methods without altering the underlying analytical model.

Following dashboard development, the operational metrics displayed within the report were validated against the corresponding SQL query results. Average delivery days, shipping costs, and shipping-mode performance were carefully verified to ensure consistency between the analytical database and the Power BI reporting environment. This validation confirmed that the Shipping and Operations Dashboard accurately represented the organization's logistics performance and provided reliable operational insights for business decision-making.

8.7 Business Insights Dashboard Development

The final dashboard developed during the Power BI implementation phase was the Business Insights Dashboard. The objective of this dashboard was to consolidate the analytical findings generated throughout the project into a single reporting interface that highlights significant business trends, comparative performance indicators, and decision-support information. Unlike the previous dashboards, which focused on individual business domains, this dashboard was designed to provide a comprehensive analytical view by combining insights obtained from sales, customer, product, regional, and operational analyses.

The development process involved selecting visualizations that effectively summarized the most important business findings identified during the SQL analysis. Visual components were arranged to emphasize relationships between different business dimensions while maintaining a clear and logical reporting structure. The dashboard was designed to enable stakeholders to identify performance patterns, recognise high-performing business areas, and observe factors influencing organizational profitability and operational efficiency.

During implementation, the established dashboard design standards were consistently applied. Visual formatting, colour selection, chart alignment, titles, legends, and data labels were standardized to maintain uniformity across all report pages. Interactive filtering functionality was retained to allow users to explore business insights across different customer segments, product categories, regions, and shipping methods while preserving consistency throughout the reporting environment.

After completing the dashboard development, all visual outputs were validated against the analytical results obtained during the SQL phase to ensure reporting accuracy. The verification process confirmed that the Business Insights Dashboard accurately reflected the consolidated analytical findings and successfully transformed the project's business analysis into an interactive decision-support solution.

9. Challenges and Solutions

Throughout the development of the Walmart Retail Sales Analysis project, several technical and analytical challenges were encountered during the data preprocessing, database implementation, and dashboard development phases. These challenges required systematic investigation and appropriate corrective measures to ensure the reliability, consistency, and accuracy of the final analytical solution. This chapter documents the major issues faced during project development and describes the

approaches adopted to resolve them. Recording these challenges not only demonstrates the practical aspects of project implementation but also provides valuable reference for future analytical projects involving similar datasets and technologies.

The following sections describe the challenges encountered during the Python preprocessing phase, SQL database implementation, and Power BI dashboard development, together with the solutions implemented to overcome each issue.

9.1. Python Development Challenges

The Python preprocessing phase involved transforming the raw Walmart Retail Sales dataset into a clean and analysis-ready dataset. During this stage, multiple data quality issues were identified that could potentially affect the accuracy of subsequent SQL analysis and Power BI visualizations. Each issue was systematically investigated and resolved using appropriate preprocessing techniques to ensure data consistency and reliability.

One of the primary challenges encountered was the presence of missing values within the dataset. The Customer Age attribute contained a significant number of missing records, while the Product Base Margin attribute also contained null values. Since Product Base Margin represented only a very small percentage of the overall dataset, the records containing missing values for this attribute were removed to avoid introducing unnecessary estimation bias. After eliminating these records, the remaining missing values in the Customer Age attribute were handled using median imputation, as the median is less sensitive to extreme values and provides a more representative estimate for numerical attributes containing outliers.

Another challenge involved identifying and handling duplicate records within the dataset. Duplicate observations can distort statistical summaries, influence business metrics, and negatively impact visualization accuracy. Duplicate detection procedures were therefore performed, and redundant records were removed to ensure that each transaction was uniquely represented within the analytical dataset.

The dataset also contained inconsistent categorical values and formatting irregularities across several attributes. Standardization procedures were applied to maintain consistent naming conventions, improve data readability, and eliminate formatting inconsistencies before further analysis. Date fields were converted into appropriate datetime formats, allowing chronological analysis and feature engineering to be performed accurately during subsequent stages of the project.

Outlier detection represented another important preprocessing challenge. Several numerical attributes, including Sales, Profit, and Shipping Cost, exhibited unusually high or low values. Instead of removing these observations, they were carefully examined because such extreme values may represent legitimate business transactions rather than data entry errors. Since the objective of the project was to analyse actual retail business performance, these outliers were retained to preserve the integrity and realism of the dataset.

Following the completion of preprocessing activities, the cleaned dataset was thoroughly validated to ensure that all transformations had been successfully applied without compromising data integrity. The resulting dataset served as the standardized analytical foundation for the subsequent Exploratory Data Analysis, SQL Business Analysis, and Power BI Dashboard Development phases of the project.

9.2. SQL Development Challenges

The SQL phase of the project involved importing the cleaned analytical dataset into MySQL, validating the database structure, and performing business-oriented analyses using SQL queries. During this stage, several technical challenges were encountered that affected database implementation and query execution. Each issue was systematically investigated and resolved before proceeding with business analysis.

One of the initial challenges involved importing the cleaned dataset into MySQL. The dataset could not be imported successfully using the standard import procedure because MySQL restricted local file access for security reasons. This resulted in errors while attempting to load the CSV file into the database. The issue was resolved by enabling the LOCAL INFILE configuration within MySQL and modifying the client connection settings to permit local file imports. After updating the configuration, the cleaned dataset was successfully imported into the database.

Another challenge was related to CSV formatting and data compatibility during the import process. Minor inconsistencies in file formatting affected the interpretation of certain records while loading the dataset. To resolve this issue, the cleaned dataset generated during the Python preprocessing phase was verified and exported using a consistent CSV format before being imported into MySQL. This ensured that all records and attributes were correctly mapped to their respective database columns.

Following the successful import, database validation became an essential step before commencing business analysis. Validation queries were executed to verify the total number of imported records, identify duplicate transactions, confirm unique values, and inspect missing or inconsistent data within important attributes. These verification

procedures ensured that the SQL database accurately represented the cleaned analytical dataset produced during the preprocessing phase.

During the business analysis stage, several SQL queries required refinement due to their increasing complexity. Business questions involving rankings, cumulative calculations, grouped aggregations, and comparative analysis resulted in lengthy query structures. To improve readability and maintainability, advanced SQL features such as Common Table Expressions (CTEs), window functions, ranking functions, and CASE expressions were incorporated wherever appropriate. This approach simplified complex analytical queries while maintaining accurate business results.

After resolving these implementation challenges, the SQL database became a stable analytical environment capable of supporting comprehensive business analysis. The validated SQL outputs subsequently served as the reference for verifying the Power BI dashboard, ensuring consistency across all phases of the project.

9.3. Power BI Development Challenges

The Power BI phase focused on transforming the validated analytical dataset into an interactive business intelligence dashboard. Although the dataset had already been cleaned and analysed during the Python and SQL phases, several challenges were encountered while developing an effective and reliable reporting solution. These challenges primarily involved data verification, visualization design, dashboard organization, and ensuring consistency between analytical results and dashboard outputs.

One of the primary challenges was ensuring that the imported dataset within Power BI accurately reflected the cleaned dataset generated during the preprocessing stage. Before visualization development commenced, all imported fields, data types, and record counts were verified to ensure consistency with the analytical dataset used during SQL analysis. This validation was essential to prevent discrepancies between the database results and the dashboard visualizations.

Another challenge involved selecting appropriate visualizations for presenting different business metrics. Since multiple analytical dimensions such as sales performance, customer behaviour, regional distribution, and shipping operations were being represented, considerable effort was required to determine which visualization best communicated each business insight. Different chart types were evaluated before finalizing the dashboard layout to maximize readability while avoiding unnecessary visual complexity.

Maintaining consistency across multiple dashboard pages also presented a significant design challenge. As several report pages were developed for different business domains, it was necessary to standardize formatting elements including visual alignment, colour schemes, titles, legends, labels, and spacing. Establishing uniform formatting guidelines ensured that the dashboard maintained a professional appearance and provided a consistent user experience throughout the reporting environment.

During dashboard development, an issue was encountered with the map visualization, where geographical data was initially not rendered correctly despite appropriate configuration. The issue required investigation of Power BI settings and visualization configuration before the expected geographical representation could be achieved. Resolving this issue ensured that location-based analyses could be accurately incorporated into the dashboard.

The final challenge involved validating the dashboard outputs against the analytical results generated during the SQL phase. Every major business metric presented within the dashboard was cross-verified with the corresponding SQL query results to ensure consistency and reporting accuracy. This verification process confirmed that the Power BI dashboard faithfully represented the analytical findings produced during earlier project stages and served as a reliable decision-support tool.

Project Repository

The complete project implementation, including the dataset source, Python notebooks, SQL scripts, Power BI dashboard file, and supporting resources, is available at:

GitHub Repository: github.com/ganeshsai23/Walmart-Retail-Sales-Analysis